

Experiment 8

Aim: To explain prediction of a pre-trained model using LIME.

Procedure:

1. Take any pretrained ImageNet model (VGG-16, ResNet-50, Inception v3)
2. Select three pictures, each with one clear main object, on three clearly different backgrounds.
3. Apply LIME and generate attribution maps.

Task:

1. Take ONE picture. Find two different LIME settings that both look reasonable, but which give you two explanations that disagree with each other. (Vary: number of superpixels, segmentation algorithm, number of perturbations, or random seed — one at a time.)
2. State, in advance, what result would count as evidence that explanation A is correct, and what would count as evidence for B.
3. Design and run an experiment — using only the model's predictions — to determine which of the two explanations, if either, is correct.